Data Storytelling & Statistical Validation

ApexPlanet Data Analytics Internship – Task 4

Objective: *To transform analytical findings into meaningful business insights and validate them using statistical methods.*

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Business Objective

The objective of this project is to analyse the sales dataset and identify meaningful business insights. The analysis focuses on understanding sales performance, customer behaviour, product performance, and city-wise revenue to support better business decisions.

Goals

- Analyze overall sales performance.
- Identify the highest-selling product category.
- Find the top-performing cities.
- Understand customer purchasing patterns.
- Validate findings using statistical testing.

Dataset Overview

Dataset Information

- **Source:** Apex Planet Software Pvt. Ltd.
- **Records:** 992 Sales Transactions
- **Categories:** Electronics, Fashion, Grocery, Education & Furniture

Key Attributes

- Order Details
- Product & Category
- Quantity & Total Sales
- Customer Demographics (Age, Gender)
- City

Data Preparation

- Missing values and duplicates removed
- Date format standardized
- Dataset prepared for analysis

Dashboard Summary

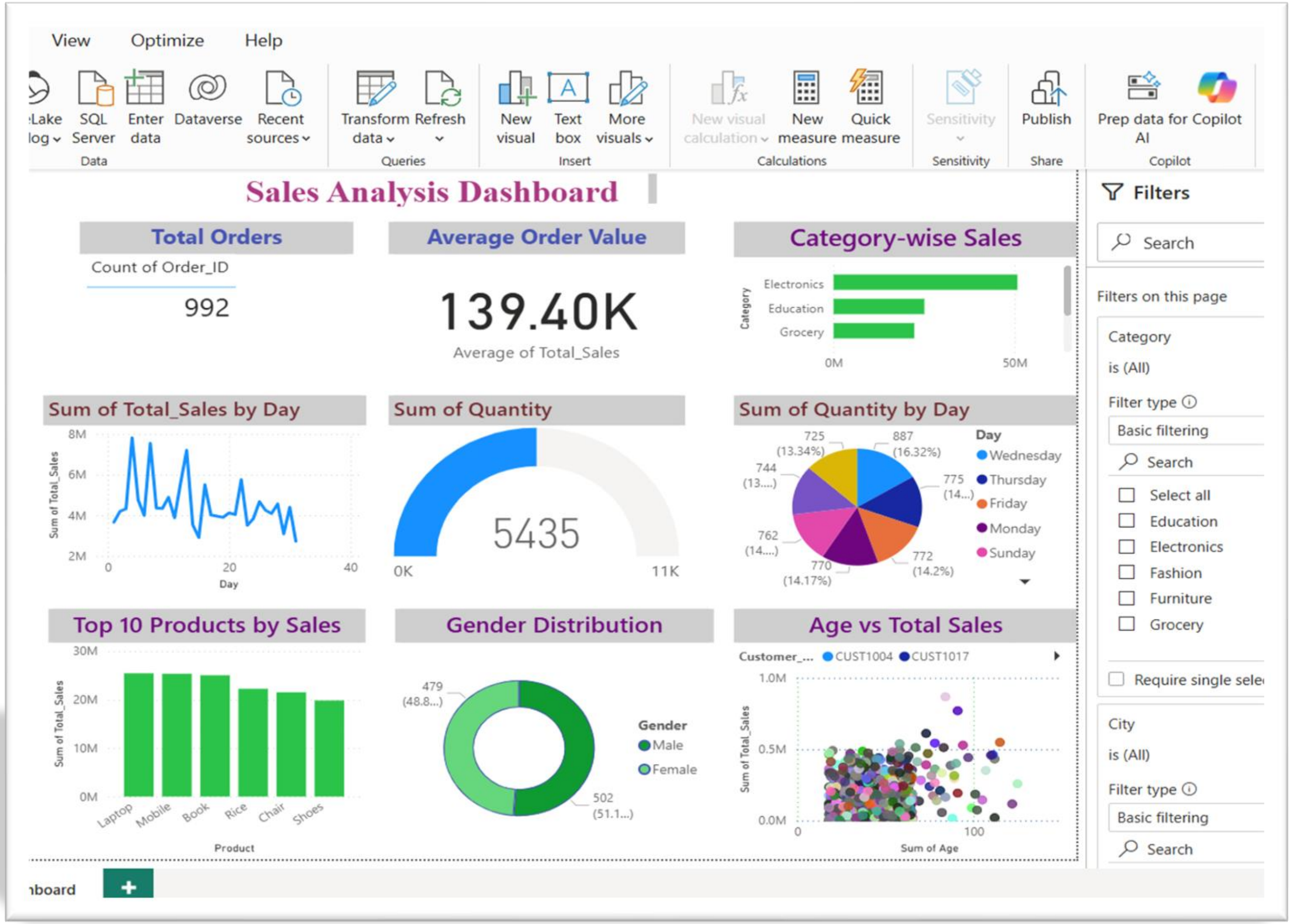
Total Revenue: ₹139,399,439.65

Total Orders: 992

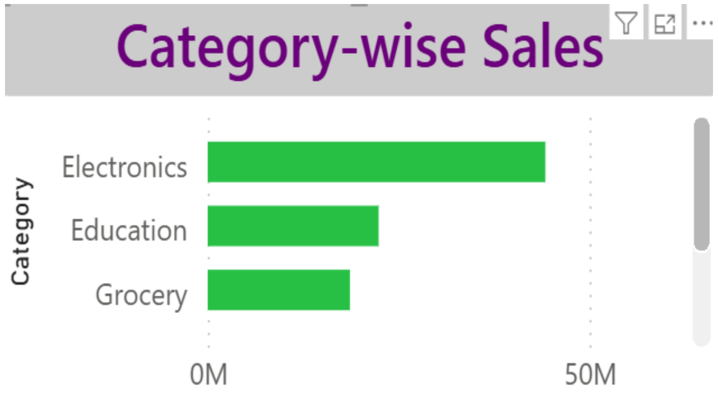
Average Order Value: ₹139,399.44

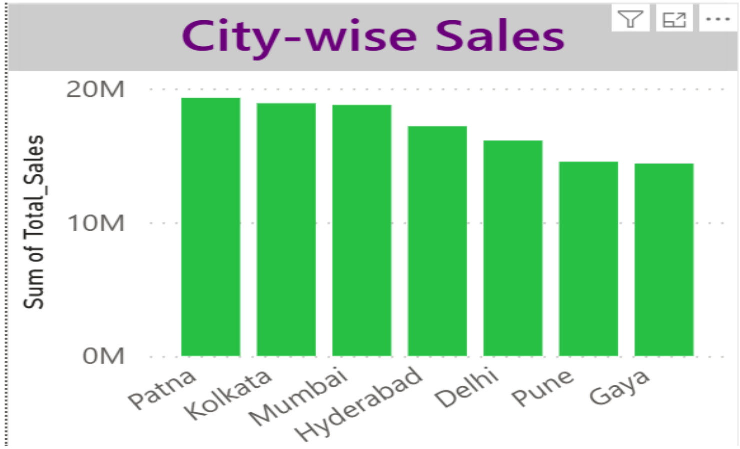
Top Category: Electronics

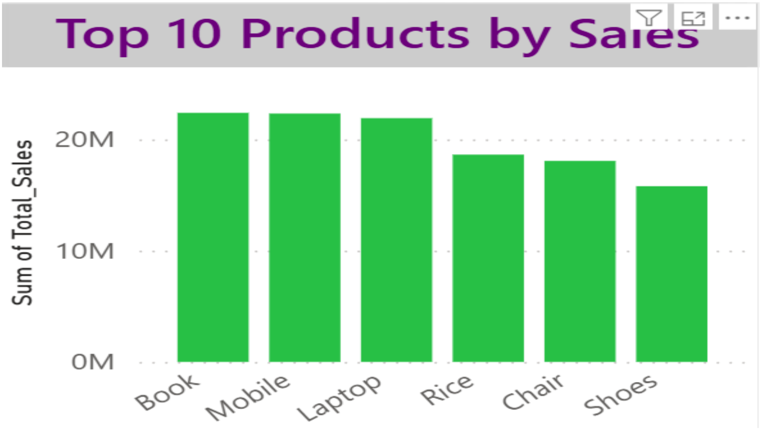
Top City: Patna



Key Business Insights



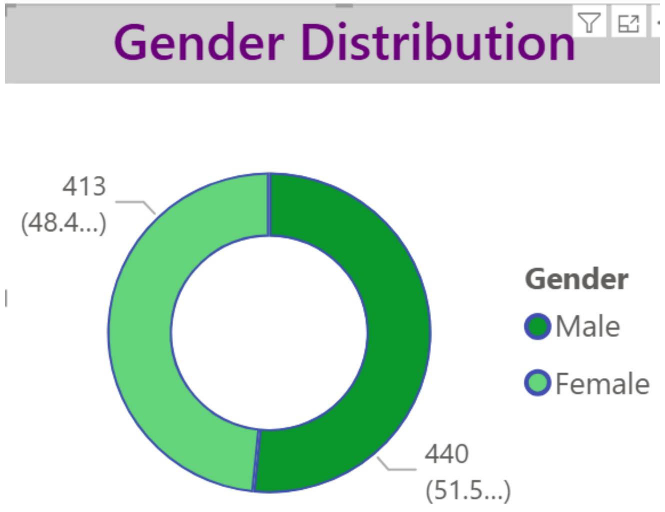


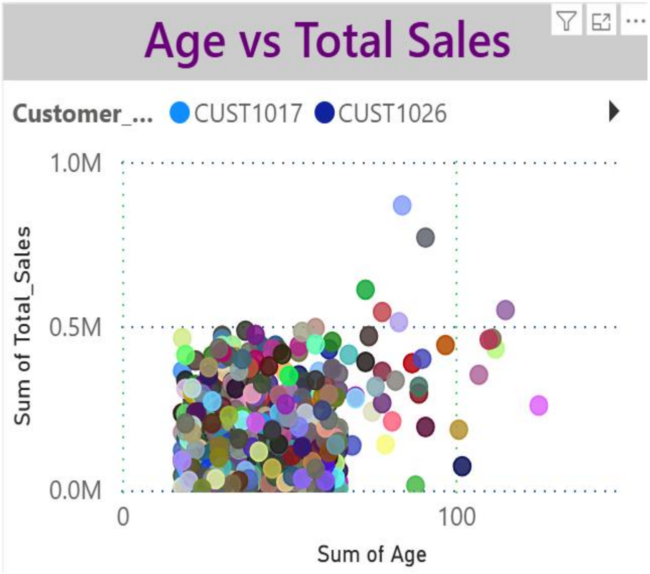


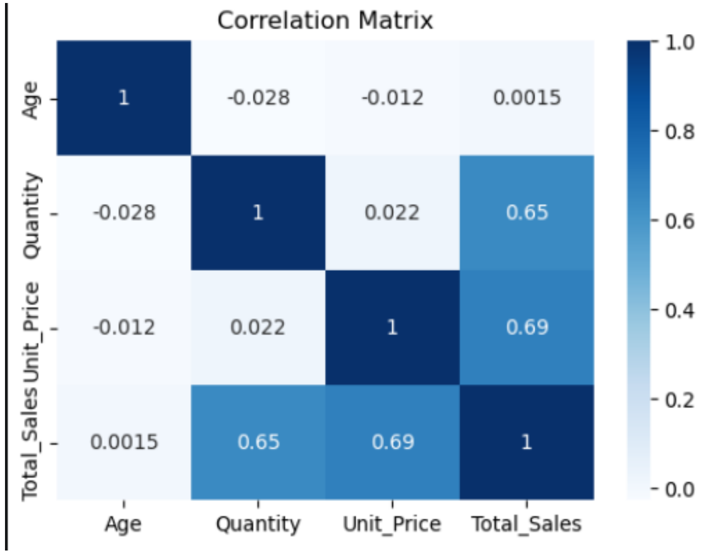
Key Findings

- Electronics generated the highest revenue among all categories.
- Patna recorded the highest sales revenue.
- Fashion contributed the lowest revenue.
- Sales are concentrated among a few high-performing products.
- Revenue varies across different cities.

Customer Analysis







Observations

- Sales are distributed across both male and female customers.
- Customer age shows only a weak relationship with sales.
- Quantity sold and Total Sales have a positive correlation.
- Most purchasing behaviour is influenced by product demand rather than customer age.

Hypothesis Testing

Null Hypothesis (H_0):

There is no significant difference in average sales across product categories.

Alternative Hypothesis (H_1):

There is a significant difference in average sales across product categories.

Test Used : One-Way ANOVA

Result

- F Statistic: **0.7171**
- P Value: **0.5803**

```python
import pandas as pd
from scipy.stats import f_oneway

df=pd.read_excel(r"C:\Users\prash\OneDrive\docs\Documents\Desktop\ApexPlanet_DataAnalytics_Dataset.xlsx")
groups = [group["Total_Sales"].values for _, group in df.groupby("Category")]

f_stat, p_value = f_oneway(*groups)

print("F Statistic:", f_stat)
print("P Value:", p_value)
```

```
F Statistic: 0.7171167721714616
P Value: 0.5802864995731378
```

Conclusion

Summary

- Successfully analysed the sales dataset using Python, SQL, and Power BI.
- Identified key performance indicators (KPIs) and business trends.
- Electronics emerged as the highest-performing product category.
- Patna generated the highest revenue among all cities.
- Statistical testing confirmed that category-wise sales differences are not statistically significant.

Business Impact

- The insights obtained from this analysis can help businesses improve inventory planning, sales strategies, and decision-making through data-driven approaches.